Random Initialization of Response Probabilities

Directives

1. Use Dirichlet(1, ..., 1) per (class, item) block

When initializing `vecprobs`, draw each (class, item) probability vector from a **Dirichlet(1, ..., 1)** distribution. This is the only method that produces a **uniform distribution over the probability simplex**.

Implement by drawing `K` independent `Gamma(1, 1)` (i.e. `Exp(1)`) variates and normalizing them to sum to 1.0.

Do not normalize independent `Uniform(0, 1)` draws. That method is biased toward the center of the simplex.

2. Respect the `vecprobs` layout

`vecprobs` is a flattened vector with layout:

```
[class 0, item 0, cat 0] ... [class 0, item 0, cat K0-1]
[class 0, item 1, cat 0] ... [class 0, item 1, cat K1-1]
...
[class R-1, last item, cat 0] ... [class R-1, last item, cat KJ-1]
```

Total length = `nclass * sum_j(num_choices[j])`.

For every `(r, j)` block of length `num_choices[j]`, the values must sum to `1.0`.

3. Make the seed configurable and use it

The initialization function must accept an `unsigned int seed` parameter and use it to seed the random number generator (e.g. `std::mt19937`). This ensures reproducible results across runs.

4. Implementation pattern

```
static Eigen::VectorXd random_init_probs(const std::vector<int>& num_choices,
                                         int nclass,
                                         unsigned int seed) {
    int total = 0;
    for (int k : num_choices) total += k;

    Eigen::VectorXd vecprobs(nclass * total);
    std::mt19937 gen(seed);
    std::gamma_distribution<double> gamma(1.0, 1.0);

    for (int r = 0; r < nclass; ++r) {
        int pos = 0;
        for (size_t j = 0; j < num_choices.size(); ++j) {
            int K = num_choices[j];
            double sum = 0.0;
            for (int k = 0; k < K; ++k) {
                double draw = gamma(gen);
                vecprobs(r * total + pos + k) = draw;
                sum += draw;
            }
            // Normalize so probabilities for this (class, item) sum to 1
            for (int k = 0; k < K; ++k) {
                vecprobs(r * total + pos + k) /= sum;
```

```
            }
            pos += K;
        }
    }
    return vecprobs;
}
```

Mathematical Justification

1. Why Dirichlet(1,...,1) is uniform on the simplex

The probability simplex in K dimensions is

$$\Delta_{K-1} = \left\{ (p_1, \dots, p_K) \,\middle|\, p_k \geq 0,\ \sum_{k=1}^{K} p_k = 1 \right\}.$$

A **uniform** distribution on this set has constant density everywhere on Δ_{K-1}. The Dirichlet distribution with parameter vector $\alpha = (\alpha_1, \dots, \alpha_K)$ has density

$$f(p_1, \dots, p_K) = \frac{1}{B(\alpha)} \prod_{k=1}^{K} p_k^{\alpha_k - 1}, \qquad B(\alpha) = \frac{\prod_{k=1}^{K} \Gamma(\alpha_k)}{\Gamma\left(\sum_{k=1}^{K} \alpha_k\right)}.$$

When $\alpha_k = 1$ for all k, the density becomes

$$f(p_1, \dots, p_K) = \frac{1}{B(1, \dots, 1)} \prod_{k=1}^{K} p_k^0 = \frac{\Gamma(K)}{\Gamma(1)^K} \cdot 1 = (K-1)!,$$

which is constant on Δ_{K-1}. Therefore $\mathrm{Dir}(1, \dots, 1)$ is exactly the uniform distribution on the simplex.

2. Proof that Gamma normalization produces Dir(1,...,1)

Let $X_1, \dots, X_K \overset{\text{iid}}{\sim} \mathrm{Gamma}(1, 1)$, i.e. the standard exponential distribution with density $f_X(x) = e^{-x}$ for $x > 0$. Define

$$S = \sum_{j=1}^{K} X_j, \qquad P_k = \frac{X_k}{S}.$$

We show that $(P_1, \dots, P_K)$ has the same distribution as $\mathrm{Dir}(1, \dots, 1)$.

Change of variables. We use the invertible map $(X_1, \dots, X_K) \mapsto (S, P_1, \dots, P_{K-1})$ defined by

$$X_k = S \cdot P_k \quad (k = 1, \dots, K-1), \qquad X_K = S \cdot \left(1 - \sum_{k=1}^{K-1} P_k\right) = S \cdot P_K.$$

The Jacobian matrix $\partial(X_1, \dots, X_K)/\partial(S, P_1, \dots, P_{K-1})$ has first column $(P_1, \dots, P_K)^T$ and for $j = 1, \dots, K-1$ the $(j+1)$-th column has S in row j, $-S$ in row K, and 0 elsewhere. Its determinant is

$$|J| = S^{K-1}.$$

Joint density of (S, P). Since the X_k are independent with density e^{-x_k}, the joint density is

$$f_X(x_1, \dots, x_K) = \exp\left(-\sum_{k=1}^{K} x_k\right) = e^{-S}.$$

By the change-of-variables formula,

$$f_{S,P}(s, p_1, \dots, p_{K-1}) = f_X(x_1, \dots, x_K) \cdot |J| = e^{-s} \cdot s^{K-1}.$$

Marginal density of P. The joint density factors into a function of s alone and a constant function of the p_k:

$$f_{S,P}(s,p) = \underbrace{s^{K-1}e^{-s}}_{\text{function of } s} \times \underbrace{1}_{\text{function of } p}.$$

Integrating over $s > 0$ yields the marginal density of P:

$$f_P(p) = \int_0^\infty s^{K-1}e^{-s}\, ds = \Gamma(K) = (K-1)!.$$

This is constant on the simplex, identical to the density of $\mathrm{Dir}(1, \dots, 1)$.

3. Proof that normalizing Uniform(0,1) is NOT uniform

Let $U_1, \dots, U_K \overset{\text{iid}}{\sim} \mathrm{Uniform}(0,1)$ with joint density $f_U(u) = 1$ on $[0,1]^K$. Define

$$S = \sum_{j=1}^{K} U_j, \qquad P_k = \frac{U_k}{S}.$$

We derive the density of $P = (P_1, \dots, P_K)$ on the simplex.

Change of variables with box constraints. Use the same map as above: $U_k = S \cdot P_k$ for $k = 1, \dots, K$, with $|J| = S^{K-1}$. The constraint $(U_1, \dots, U_K) \in [0,1]^K$ translates to

$$0 \le S \cdot P_k \le 1 \quad \text{for all } k \quad \iff \quad 0 \le S \le \frac{1}{\max_k P_k}.$$

Because the uniform density is 1 on the cube and 0 outside, the joint density of (S, P) is

$$f_{S,P}(s,p) = s^{K-1} \quad \text{for } 0 < s < \frac{1}{\max_k p_k},$$

and 0 otherwise.

Marginal density of P. Integrate over the allowed range of s:

$$f_P(p) = \int_0^{1/\max_k p_k} s^{K-1}\, ds = \left[\frac{s^K}{K}\right]_0^{1/\max_k p_k} = \frac{1}{K \cdot (\max_k p_k)^K}.$$

Interpretation. The density is **not** constant. It depends on $\max_k p_k$:

- At the center $p = (1/K, \dots, 1/K)$:
$$f_P(p) = \frac{K^{K-1}}{K} = K^{K-1}.$$

- At a corner, e.g. $p = (1, 0, \dots, 0)$:
$$f_P(p) = \frac{1}{K \cdot 1^K} = \frac{1}{K}.$$

The ratio of density at the center to a corner is K^K. For $K = 2$ the center is $4\times$ more likely than the corners; for $K = 3$ it is $27\times$ more likely. The distribution is **strongly biased toward the center** of the simplex.

Explicit formula for K = 2. For a 2-category item with $P_1 = p$ and $P_2 = 1 - p$, we have $\max(p, 1 - p)$ and therefore

$$f_P(p) = \frac{1}{2 \cdot \max(p, 1-p)^2} = \begin{cases} \dfrac{1}{2(1-p)^2}, & p \leq \frac{1}{2}, \\ \dfrac{1}{2p^2}, & p > \frac{1}{2}. \end{cases}$$

This peaks at $p = \frac{1}{2}$ with value 2 and decreases to $\frac{1}{2}$ at the boundaries $p = 0$ and $p = 1$.

References

1. **Devroye, L.** (1986). *Non-Uniform Random Variate Generation.* Springer, Chapter V, Section 3. Devroye derives the Dirichlet via Gamma normalization and notes that $\mathrm{Dir}(1, \dots, 1)$ is uniform on the simplex.
2. **Wikipedia**, "Dirichlet distribution", *Random variate generation* and *Special cases* sections. https://en.wikipedia.org/wiki/Dirichlet_distribution